

KRASNYE BAKI, Russian Federation

WMO: 273690

Lat: 57.133N

Lon: 45.167E

Elev: 362

StdP: 14.50

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-12.4	-23.2	1.5	-18.1	-17.2	2.2	-12.1	14.6	20.0	13.5	20.9	2.2	320	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	85.5	68.8	82.4	67.6	79.2	65.7	71.1	82.1	69.3	79.3	67.5	76.3	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.8	76.6	65.6	95.8	74.4	64.0	90.3	72.7	35.1	82.3	33.6	79.3	32.1	76.4	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.1	11.0	-24.2	89.7	7.1	3.5	-29.3	92.2	-33.5	94.2	-37.5	96.2	-42.7	98.7		
(5)			-24.4	74.1	7.1	2.3	-29.5	75.7	-33.7	77.0	-37.6	78.3	-42.8	79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	13.8	15.5	26.9	41.7	54.8	62.2	66.7	62.5	52.2	39.9	27.5	18.5		
	DBStd	20.97	13.44	12.84	8.69	8.87	9.20	7.84	6.46	7.17	7.46	7.80	10.08	12.92		
	HDD50	5180	1122	966	716	276	56	6	0	2	59	323	676	978		
	HDD65	9257	1587	1386	1181	698	334	144	58	135	387	778	1126	1443		
	CDD50	1650	0	0	0	29	206	371	518	390	126	10	0	0		
	CDD65	252	0	0	0	0	19	60	111	58	4	0	0	0		
	CDH74	2169	0	0	0	3	205	546	935	446	34	0	0	0		
	CDH80	564	0	0	0	0	44	142	254	121	3	0	0	0		
	WSAvg	5.4	5.4	5.7	6.0	5.6	5.8	5.3	4.7	4.9	5.1	5.7	5.6	5.7		
	PrecAvg	25.7	1.9	1.4	1.5	1.4	1.9	3.1	2.7	2.9	2.3	2.6	2.1	2.0		
	PrecMax	35.9	3.7	2.4	3.6	3.9	4.1	6.9	6.5	6.4	4.8	7.1	4.2	3.9		
(15) Precipitation	PrecMin	19.2	0.5	0.3	0.2	0.4	0.6	0.8	0.2	0.0	0.6	0.6	0.3	0.6		
	PrecStd	3.4	0.7	0.6	0.8	0.9	0.8	1.5	1.5	1.7	1.1	1.5	0.9	0.8		
	DB	35.9	37.9	52.8	73.0	84.9	89.1	89.9	89.0	78.1	64.6	50.1	42.2			
	MCWB	34.5	34.6	43.1	56.2	65.1	69.5	70.6	72.0	63.2	54.9	47.6	41.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	34.2	35.5	45.0	67.2	79.5	84.1	85.8	83.1	73.1	57.6	44.9	37.8			
	2% MCWB	33.0	33.5	37.7	53.1	62.1	67.3	69.8	68.7	61.1	51.5	43.0	36.4			
	5% DB	33.1	33.9	41.1	61.3	75.0	80.2	82.7	78.7	68.6	53.7	41.8	34.9			
	5% MCWB	32.0	32.5	35.3	50.1	59.1	66.0	68.4	66.3	59.1	49.8	40.2	33.7			
	10% DB	31.3	32.4	38.0	56.8	70.8	76.6	79.6	75.0	64.3	50.8	38.8	33.3			
	10% MCWB	30.2	31.1	33.9	47.2	57.2	63.8	66.9	64.5	57.2	47.8	37.6	32.3			
	0.4% WB	35.0	35.6	43.9	57.2	68.2	71.8	73.9	74.9	65.3	56.1	47.5	41.7			
	0.4% MCDB	35.6	37.3	52.1	69.5	80.6	83.2	84.8	88.2	74.0	62.1	49.3	42.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.3	34.0	39.0	54.0	64.2	69.7	71.5	70.4	62.3	52.7	43.7	36.5			
	2% MCDB	33.9	35.0	43.2	65.2	76.1	81.4	83.1	79.8	70.3	56.4	44.7	37.5			
	5% WB	32.1	32.7	36.4	51.0	61.4	67.6	69.8	67.9	59.9	50.3	40.4	33.9			
	5% MCDB	33.0	33.9	39.7	60.2	72.0	77.6	79.8	75.8	67.1	53.2	41.7	34.5			
	10% WB	30.1	31.1	34.4	47.8	58.9	65.4	68.0	65.6	57.7	48.0	37.6	32.4			
	10% MCDB	31.3	32.5	37.4	55.4	67.9	73.9	76.9	73.0	63.5	50.6	38.6	33.2			
	MDBR	10.2	12.8	14.0	16.8	18.6	17.7	17.2	16.5	14.6	10.0	7.1	8.4			
	5% DB	7.1	8.7	16.7	23.9	24.7	23.1	22.4	22.1	22.0	16.2	9.4	5.9			
(36) Mean Daily Temperature Range	MCWBR	7.2	7.7	11.2	13.3	10.8	9.8	9.2	10.0	11.7	10.8	8.2	5.9			
	MCDBR	7.5	7.9	14.0	22.7	22.2	20.7	20.3	20.1	19.9	14.5	8.4	5.6			
	MCWBR	7.7	7.3	10.1	13.1	10.5	10.0	9.5	10.3	11.3	10.6	7.8	5.7			
	MCDBR	7.7	7.3	10.1	13.1	10.5	10.0	9.5	10.3	11.3	10.6	7.8	5.7			
(40) Clear-Sky Solar Irradiance	taub	0.282	0.316	0.351	0.402	0.377	0.373	0.404	0.417	0.365	0.348	0.309	0.277			
	taud	2.071	2.053	2.037	2.128	2.365	2.407	2.332	2.274	2.401	2.333	2.198	2.045			
	Ebn at Noon	177	222	247	255	273	275	262	249	247	215	169	141			
	Edh at Noon	18	29	39	42	35	34	36	36	27	23	16	14			
(44) All-Sky Solar Radiation	RadAvg	159	404	859	1258	1693	1776	1762	1375	816	364	135	87			
	RadStd	26	94	95	130	110	148	162	141	108	63	23	22			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-333	N/A	N/A
(47) Regional (7 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	-270	-353	N/A

Nomenclature: See separate page